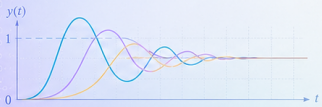
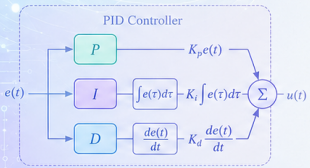
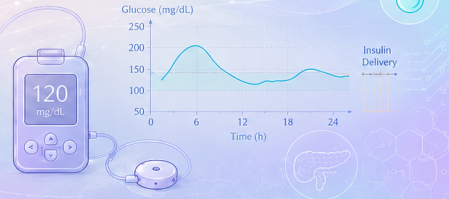
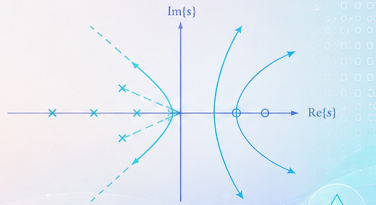
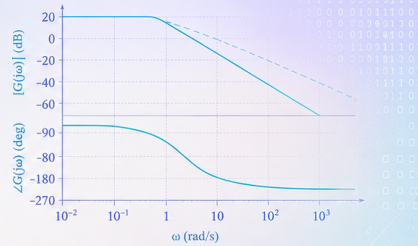


Stability

SAHAJ SAXENA
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$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \dots + b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



Stability: Basic Idea

The total response of an LTI system is

$$c(t) = c_{\text{forced}}(t) + c_{\text{natural}}(t)$$

Forced Response

Produced by the external input.

$$c_{\text{forced}}(t)$$

It usually determines the desired steady-state behaviour.

Natural Response

Produced by system dynamics and initial conditions.

$$c_{\text{natural}}(t)$$

Its behaviour determines **stability**.

For a stable system,

$$c_{\text{natural}}(t) \rightarrow 0 \quad \text{as } t \rightarrow \infty$$

Stable, Unstable, and Marginally Stable Systems

Stable

$$c_n(t) \rightarrow 0$$

The natural response decays with time.

$$e^{-at}, \quad a > 0$$

Unstable

$$|c_n(t)| \rightarrow \infty$$

The natural response grows without bound.

$$e^{at}, \quad a > 0$$

Marginally Stable

The natural response neither decays nor grows.

$$\sin(\omega t), \quad \cos(\omega t)$$

It remains bounded and oscillatory.

BIBO Stability

BIBO means

Bounded Input $\Rightarrow$ Bounded Output
--

Stable

If every bounded input satisfies

$$|u(t)| \leq M_u < \infty,$$

and produces

$$|y(t)| \leq M_y < \infty,$$

then the system is **BIBO stable**.

Unstable

If there exists a bounded input for which

$$|u(t)| \leq M_u,$$

but

$$|y(t)| \rightarrow \infty,$$

then the system is **unstable**.

Closed-Loop Poles and Stability

For an LTI system, stability can be determined from the locations of its **closed-loop poles**.

Stable System

All poles lie in the open left-half plane:

$$\boxed{\operatorname{Re}(p_i) < 0}$$

Typical natural responses are

$$e^{-at}, \quad e^{-at} \sin(\omega t).$$

Both decay to zero.

Unstable System

At least one pole lies in the right-half plane:

$$\boxed{\operatorname{Re}(p_i) > 0}$$

Typical natural responses are

$$e^{at}, \quad e^{at} \sin(\omega t).$$

Their magnitude increases with time.

Imaginary-Axis Poles

Simple Imaginary-Axis Poles

For poles

$$p_{1,2} = \pm j\omega,$$

the natural response contains

$$\sin(\omega t), \quad \cos(\omega t).$$

The amplitude remains constant.

$\Rightarrow$ *Marginally Stable*

Repeated Imaginary-Axis Poles

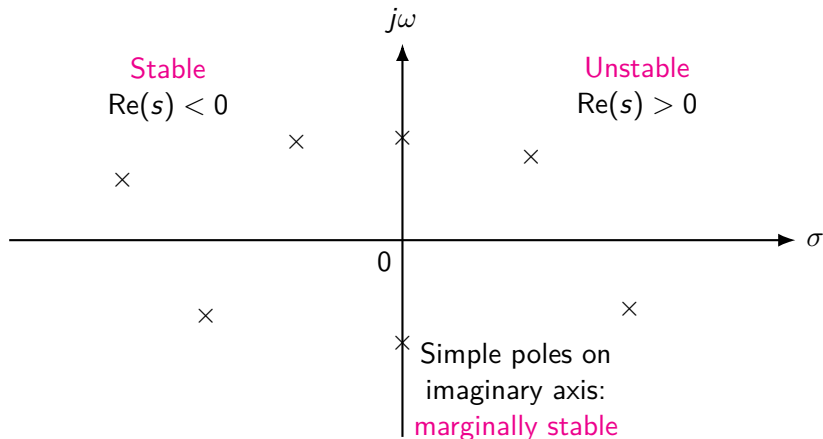
Repeated poles generate terms such as

$$t \sin(\omega t), \quad t \cos(\omega t).$$

Their amplitude increases with time.

$\Rightarrow$ *Unstable*

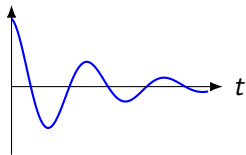
Stability Regions in the s-Plane



Natural Response and Stability

Stable

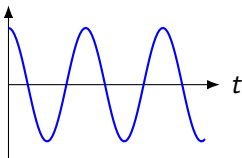
$$p = -a, \quad a > 0$$



$$c_n(t) \rightarrow 0$$

Marginally Stable

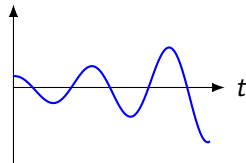
$$p = \pm j\omega$$



$$|c_n(t)| < \infty$$

Unstable

$$p = +a, \quad a > 0$$



$$|c_n(t)| \rightarrow \infty$$

Example: Stable Closed-Loop System

For the unity-feedback system

$$G(s) = \frac{3}{s(s+1)(s+2)},$$

the closed-loop transfer function is

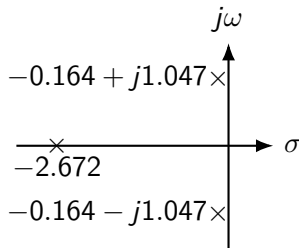
$$T(s) = \frac{3}{s^3 + 3s^2 + 2s + 3}.$$

The closed-loop poles are approximately

$$p_1 = -2.672, \quad p_{2,3} = -0.164 \pm j1.047.$$

Since $[\operatorname{Re}(p_i) < 0]$, all poles lie in the **left-half plane**.

System is stable



Example: Unstable Closed-Loop System

For the unity-feedback system

$$G(s) = \frac{7}{s(s+1)(s+2)},$$

the closed-loop transfer function is

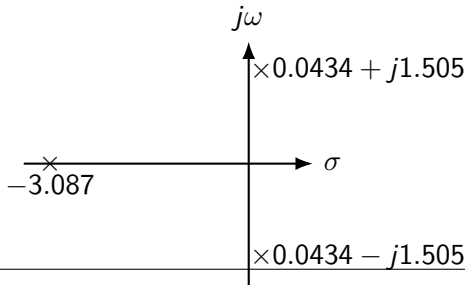
$$T(s) = \frac{7}{s^3 + 3s^2 + 2s + 7}.$$

The closed-loop poles are approximately

$$p_1 = -3.087, \quad p_{2,3} = 0.0434 \pm j1.505.$$

Since the complex poles satisfy $\text{Re}(p_{2,3}) > 0$, they lie in the **right-half plane**.

System is unstable



Stability Summary

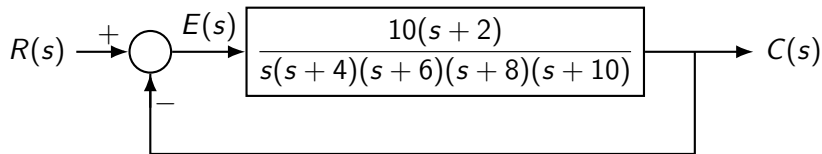
Pole Location	Natural Response	Stability
$\text{Re}(p) < 0$	Decays to zero	Stable
$\text{Re}(p) > 0$	Grows without bound	Unstable
Simple $p = \pm j\omega$	Sustained oscillation	Marginally stable
Repeated $p = \pm j\omega$	Growing oscillation	Unstable

Important

For a rational continuous-time LTI system to be **BIBO stable**, all poles must lie strictly in the open left-half plane.

Determining Stability of a Feedback System

It is not always straightforward to determine whether a feedback control system is stable.



- ▶ The poles of the forward transfer function $G(s)$ are easy to identify.
- ▶ However, **stability depends on the closed-loop poles**.
- ▶ These poles are obtained from the closed-loop characteristic equation

$$1 + G(s) = 0.$$

Equivalent Closed-Loop System

For unity negative feedback,

$$T(s) = \frac{G(s)}{1 + G(s)}.$$

For

$$G(s) = \frac{10(s + 2)}{s(s + 4)(s + 6)(s + 8)(s + 10)},$$

we obtain

$$T(s) = \frac{10(s + 2)}{s(s + 4)(s + 6)(s + 8)(s + 10) + 10(s + 2)}.$$

Equivalent Closed-Loop System

For unity negative feedback,

$$T(s) = \frac{G(s)}{1 + G(s)}.$$

For

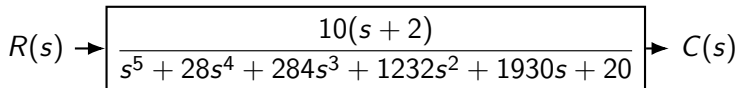
$$G(s) = \frac{10(s+2)}{s(s+4)(s+6)(s+8)(s+10)},$$

we obtain

$$T(s) = \frac{10(s+2)}{s(s+4)(s+6)(s+8)(s+10) + 10(s+2)}.$$

Expanding the denominator,

$$T(s) = \frac{10(s+2)}{s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20}$$



Why Direct Pole Calculation Becomes Difficult

The closed-loop poles are the roots of

$$s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20 = 0$$

- ▶ This is a **fifth-order characteristic equation**.
- ▶ Factoring such a polynomial by hand may be difficult.
- ▶ Numerical software can calculate the roots, but we would like a method that can determine stability **without explicitly solving for every pole**.

Question

Can we obtain useful information about stability directly from the **coefficients of the characteristic polynomial**?

A Preliminary Test from Polynomial Coefficients

Consider the characteristic polynomial

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0.$$

For a Stable System

If all poles lie in the left-half plane, the denominator can be formed from factors such as

$$(s + a_i), \quad a_i > 0,$$

and second-order factors associated with complex conjugate poles having negative real parts.

Therefore, the resulting characteristic polynomial has coefficients with the **same sign**.

Necessary Conditions

For asymptotic stability,

- ▶ all coefficients must have the **same sign**;
- ▶ no power of s should be **missing**.

If either condition is violated, the system is

unstable or, at best, marginally stable

Are Positive Coefficients Sufficient?

Consider again

$$P(s) = s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20.$$

All coefficients are positive:

$$1, \quad 28, \quad 284, \quad 1232, \quad 1930, \quad 20 > 0.$$

Also, no power of s is missing.

Are Positive Coefficients Sufficient?

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$$P(s) = s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20.$$

All coefficients are positive:

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Can we conclude that the system is stable?

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Also, no power of s is missing.

Can we conclude that the system is stable?

NO

Positive coefficients and no missing powers are **necessary conditions**, but they are not sufficient to guarantee that all poles lie in the left-half plane.

Are Positive Coefficients Sufficient?

Consider again

$$P(s) = s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20.$$

All coefficients are positive:

$$1, \quad 28, \quad 284, \quad 1232, \quad 1930, \quad 20 > 0.$$

Also, no power of s is missing.

Can we conclude that the system is stable?

NO

Positive coefficients and no missing powers are **necessary conditions**, but they are not sufficient to guarantee that all poles lie in the left-half plane.

Therefore, we need a systematic method that determines

the number of characteristic roots in the right-half plane

without explicitly calculating all roots.

Motivation for the Routh–Hurwitz Criterion

For a characteristic polynomial

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0,$$

we would like to determine stability using only its coefficients.

- ▶ Direct root calculation may become difficult for high-order systems.
- ▶ Inspection of coefficient signs provides only a **necessary condition**.
- ▶ We need to determine whether any roots lie in the **right-half s -plane**.
- ▶ This can be accomplished without explicitly finding the roots.

Next Topic

Routh–Hurwitz Stability Criterion

Routh–Hurwitz Stability Criterion

Consider the characteristic polynomial of a closed-loop system:

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0.$$

- ▶ The system is stable if all roots of $P(s) = 0$ lie in the **left-half plane**.
- ▶ Explicitly calculating the roots may be difficult for a high-order polynomial.
- ▶ The **Routh–Hurwitz criterion** determines stability directly from the coefficients of $P(s)$.

Main Result

The number of sign changes in the **first column of the Routh array** equals the number of roots of $P(s)$ in the right-half plane.

Objective of the Routh Criterion

For

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0,$$

the Routh criterion answers:

How many roots lie in the right-half s -plane?

Stable

No roots in the right-half plane:

$$N_{\text{RHP}} = 0.$$

Hence, there are no sign changes in the first column.

Unstable

If the first column contains sign changes,

$$N_{\text{RHP}} > 0,$$

the system has right-half-plane poles and is unstable.

Construction of the Routh Array

Consider

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \cdots + a_1 s + a_0.$$

The first two rows are formed directly from the coefficients:

$$\begin{array}{c|cccc} s^n & a_n & a_{n-2} & a_{n-4} & \cdots \\ s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} & \cdots \\ s^{n-2} & b_1 & b_2 & b_3 & \cdots \\ s^{n-3} & c_1 & c_2 & c_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \\ s^0 & \cdots & & & \end{array}$$

Important

The coefficients are arranged in descending powers of s , alternating between the first and second rows.

Calculating the Routh Array Elements

The elements of the third row are obtained from the first two rows.

$$b_1 = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}}$$

$$b_2 = \frac{a_{n-1}a_{n-4} - a_n a_{n-5}}{a_{n-1}}$$

Similarly,

$$c_1 = \frac{b_1 a_{n-3} - a_{n-1} b_2}{b_1}.$$

Procedure

Continue the same operation row by row until the s^0 row is obtained.

Stability Condition from the Routh Array

After constructing the Routh array, inspect only its **first column**.

$$\begin{array}{c|c} s^n & r_1 \\ s^{n-1} & r_2 \\ s^{n-2} & r_3 \\ s^{n-3} & r_4 \\ \vdots & \vdots \\ s^0 & r_{n+1} \end{array}$$

- The number of **sign changes** in

$$r_1, r_2, r_3, \dots, r_{n+1}$$

equals the number of right-half-plane roots.

- For an asymptotically stable system, all entries in the first column must have the **same sign**.

No sign change $\Rightarrow$ no RHP poles

Example 1: Stable System

Consider

$$P(s) = s^3 + 2s^2 + 3s + 4.$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & 3 \\ s^2 & 2 & 4 \\ s^1 & \frac{2(3) - 1(4)}{2} & 0 \\ s^0 & 4 & \end{array} \Rightarrow \begin{array}{c|cc} s^3 & 1 & 3 \\ s^2 & 2 & 4 \\ s^1 & 1 & 0 \\ s^0 & 4 & \end{array}$$

First Column

1, 2, 1, 4

There is **no sign change**.

System is stable

Example 2: Unstable System

Consider

$$P(s) = s^3 + 2s^2 + s + 4.$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & 2 & 4 \\ s^1 & \frac{2(1) - 1(4)}{2} & 0 \\ s^0 & 2 & 4 \end{array} \Rightarrow \begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & 2 & 4 \\ s^1 & -1 & 0 \\ s^0 & 4 & \end{array}$$

The first column is

$$1, \quad 2, \quad -1, \quad 4.$$

There are two sign changes:

$$+ \rightarrow - \quad \text{and} \quad - \rightarrow +.$$

Two RHP poles $\Rightarrow$ system is unstable
--

Stability Range of a Parameter

Consider

$$P(s) = s^3 + 3s^2 + (K + 2)s + 4K.$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & K + 2 \\ s^2 & 3 & 4K \\ s^1 & \frac{3(K + 2) - 4K}{3} & 0 \\ s^0 & 4K & \end{array}$$

Therefore,

$$s^1 : \quad \frac{6 - K}{3}.$$

For stability, every first-column entry must be positive:

$$1 > 0, \quad 3 > 0, \quad \frac{6 - K}{3} > 0, \quad 4K > 0.$$

Hence, $0 < K < 6$,

Special Case I: Zero in the First Column

During construction of the Routh array, the first element of a row may become zero while other elements in that row are nonzero. For example,

$$\begin{array}{c|ccc} s^n & \dots & \dots & \dots \\ s^{n-1} & 0 & a & b \end{array}$$

Direct calculation of the next row is then impossible because division by zero occurs.

Remedy: ε -Method

Replace the zero by a small positive number

$$0 \rightarrow \varepsilon, \quad \varepsilon > 0,$$

complete the Routh array, and finally examine the signs as

$$\varepsilon \rightarrow 0^+.$$

Special Case II: Entire Row of Zeros

Another special case occurs when an entire row becomes zero:

$$\begin{array}{c|ccc} s^{k+1} & a & b & c \\ s^k & 0 & 0 & 0 \end{array}$$

- ▶ Form an **auxiliary polynomial** using the coefficients of the row immediately above the zero row.

- ▶ Suppose

$$A(s) = as^{k+1} + bs^{k-1} + cs^{k-3} + \dots$$

- ▶ Differentiate it:

$$\frac{dA(s)}{ds}.$$

- ▶ Replace the zero row by the coefficients of $\frac{dA(s)}{ds}$, then continue the Routh array normally.

Meaning of an Entire Row of Zeros

An entire row of zeros indicates that the characteristic polynomial contains roots that are **symmetrically located about the origin**.

Possible symmetric roots include

$$\pm a, \quad \pm j\omega, \quad \pm\sigma \pm j\omega.$$

Important

A row of zeros does not by itself imply stability.

The auxiliary polynomial must be examined to determine the locations of the associated roots.

For simple imaginary-axis roots,

$$s = \pm j\omega,$$

the system may exhibit sustained oscillations and be marginally stable, provided all remaining poles lie in the left-half plane.

Example: Routh Table with a Row of Zeros

Consider the closed-loop transfer function

$$T(s) = \frac{128}{s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128}.$$

Problem

Find the number of poles in the

left-half plane, right-half plane, $j\omega$ -axis,

and conclude the stability of the closed-loop system.

Example: Routh Table with a Row of Zeros

Consider the closed-loop transfer function

$$T(s) = \frac{128}{s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128}.$$

Problem

Find the number of poles in the

left-half plane, right-half plane, $j\omega$ -axis,

and conclude the stability of the closed-loop system.

The characteristic equation is obtained from the denominator:

$$s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128 = 0$$

Initial Routh Table

For

$$P(s) = s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128,$$

the first two rows are

$$\begin{array}{c|ccccc} s^8 & 1 & 10 & 48 & 128 & 128 \\ s^7 & 3 & 24 & 96 & 192 & 0 \end{array}$$

Initial Routh Table

For

$$P(s) = s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128,$$

the first two rows are

$$\begin{array}{c|ccccc} s^8 & 1 & 10 & 48 & 128 & 128 \\ s^7 & 3 & 24 & 96 & 192 & 0 \end{array}$$

Computing the next row,

$$s^6 : \quad 2 \quad 16 \quad 64 \quad 128$$

or, dividing by 2,

$$s^6 : \quad 1 \quad 8 \quad 32 \quad 64$$

Initial Routh Table

For

$$P(s) = s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128,$$

the first two rows are

$$\begin{array}{c|ccccc} s^8 & 1 & 10 & 48 & 128 & 128 \\ s^7 & 3 & 24 & 96 & 192 & 0 \end{array}$$

Computing the next row,

$$s^6 : \quad 2 \quad 16 \quad 64 \quad 128$$

or, dividing by 2,

$$s^6 : \quad 1 \quad 8 \quad 32 \quad 64$$

The next row becomes

$$s^5 : \quad 0 \quad 0 \quad 0 \quad 0$$

Important

A complete row of zeros appears at the s^5 row. Therefore, we must use the **auxiliary polynomial**.

Auxiliary Polynomial and Completion of the Routh Table

The row above the zero row is the s^6 row:

1 8 32 64

Hence, the auxiliary polynomial is

$$A(s) = s^6 + 8s^4 + 32s^2 + 64$$

Auxiliary Polynomial and Completion of the Routh Table

The row above the zero row is the s^6 row:

$$1 \quad 8 \quad 32 \quad 64$$

Hence, the auxiliary polynomial is

$$A(s) = s^6 + 8s^4 + 32s^2 + 64$$

Differentiate:

$$\frac{dA(s)}{ds} = 6s^5 + 32s^3 + 64s$$

Thus, replace the zero row by the coefficients of $A'(s)$:

$$s^5 : \quad 6 \quad 32 \quad 64 \quad 0$$

For convenience, divide by 2:

$$s^5 : \quad 3 \quad 16 \quad 32 \quad 0$$

Now complete the table:

s^8	1	10	48	128	128
s^7	3	24	96	192	0
s^6	1	8	32	64	
s^5	3	16	32	0	
s^4	1	8	24		
s^3	-1	-5	0		
s^2	1	8			
s^1	3				
s^0	8				

Inspect the Routh table carefully.

Step 1: Even polynomial

The auxiliary polynomial

$$A(s) = s^6 + 8s^4 + 32s^2 + 64$$

is of sixth order.

From the s^6 row downward, the first column is

$$1, \quad 3, \quad 1, \quad -1, \quad 1, \quad 3, \quad 8$$

There are **two sign changes**.

Hence, the even polynomial contributes

2 RHP poles

By symmetry, it must also contribute

2 LHP poles

Since it is sixth order, the remaining poles must be

2 on the $j\omega$ -axis

Step 2: Remaining part

From the beginning of the table down to the s^6 row, the first column is

$$1, \quad 3, \quad 1$$

There is **no sign change**. Thus, the remaining second-order factor contributes

0 RHP poles

and therefore

2 LHP poles

Final Count

RHP poles	=	2
LHP poles	=	4
$j\omega$ -axis poles	=	2

The closed-loop system is unstable.

Poles of a State-Space System

Consider the LTI state-space model

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + B\mathbf{u}(t), \quad \mathbf{y}(t) = C\mathbf{x}(t) + D\mathbf{u}(t).$$

The poles of the system are determined by the **state matrix** A .

They are the values of s satisfying

$$\det(sI - A) = 0$$

This equation is called the **characteristic equation**.

Equivalently, the poles are the

eigenvalues of A

because eigenvalues satisfy

$$\det(\lambda I - A) = 0.$$

Hence,

$$p_i = \lambda_i(A)$$

Example: Finding Poles from the State Matrix

Consider

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}.$$

Form

$$sI - A = \begin{bmatrix} s & -1 \\ 6 & s + 5 \end{bmatrix}.$$

The characteristic equation is

$$\det(sI - A) = 0.$$

Therefore,

$$\begin{vmatrix} s & -1 \\ 6 & s + 5 \end{vmatrix} = 0.$$

Expanding the determinant,

$$s(s + 5) + 6 = 0.$$

Thus,

$$s^2 + 5s + 6 = 0.$$

Factorizing,

$$(s + 2)(s + 3) = 0.$$

Hence,

$$\boxed{p_1 = -2, \quad p_2 = -3}$$

Both poles lie in the left-half plane.

State-Space Poles and Transfer-Function Poles

For the state-space model,

$$G(s) = C(sI - A)^{-1}B + D.$$

Using

$$(sI - A)^{-1} = \frac{\text{adj}(sI - A)}{\det(sI - A)},$$

we obtain

$$G(s) = C \frac{\text{adj}(sI - A)}{\det(sI - A)} B + D.$$

Key Connection

The roots of

$$\det(sI - A) = 0$$

are the eigenvalues of A and determine the internal system modes.

For a minimal realization, these are also the poles of $G(s)$.

State-Space Poles and Stability

For

$$\dot{\mathbf{x}} = A\mathbf{x},$$

the natural response is governed by

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0).$$

Therefore, the eigenvalues of A determine the behaviour of the natural response.

Stable

$$\operatorname{Re}(\lambda_i) < 0$$

for all i .

Natural response decays.

Unstable

At least one

$$\operatorname{Re}(\lambda_i) > 0.$$

Natural response grows.

Marginal Case

Simple eigenvalues on the imaginary axis:

$$\operatorname{Re}(\lambda_i) = 0.$$

Sustained oscillations may occur.

Example: Stability from State-Space Model

Consider the system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 3 & 1 \\ 2 & 8 & 1 \\ -10 & -5 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u, \quad y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}.$$

Problem

Determine how many poles lie in the

left-half plane, right-half plane, $j\omega$ -axis.

Example: Stability from State-Space Model

Consider the system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 3 & 1 \\ 2 & 8 & 1 \\ -10 & -5 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u, \quad y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}.$$

Problem

Determine how many poles lie in the

left-half plane, right-half plane, $j\omega$ -axis.

The poles of a state-space system are obtained from

$$\boxed{\det(sI - A) = 0}.$$

Characteristic Equation

First form

$$sI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 3 & 1 \\ 2 & 8 & 1 \\ -10 & -5 & -2 \end{bmatrix}.$$

Therefore,

$$sI - A = \begin{bmatrix} s & -3 & -1 \\ -2 & s-8 & -1 \\ 10 & 5 & s+2 \end{bmatrix}$$

Now evaluate

$$\det(sI - A).$$

Expanding the determinant gives

$$\det(sI - A) = s^3 - 6s^2 - 7s - 52.$$

Hence, the characteristic equation is

$$s^3 - 6s^2 - 7s - 52 = 0$$

Instead of explicitly solving this cubic equation, we can use the **Routh-Hurwitz criterion**.

Routh Table and Pole Distribution

For

$$P(s) = s^3 - 6s^2 - 7s - 52,$$

the first two rows of the Routh array are

$$\begin{array}{c|cc} s^3 & 1 & -7 \\ s^2 & -6 & -52 \end{array}$$

The s^1 element is

$$b_1 = \frac{(-6)(-7) - (1)(-52)}{-6} = -\frac{47}{3}.$$

Therefore,

s^3	1	-7
s^2	-6	-52
s^1	$-\frac{47}{3}$	0
s^0	-52	

The first column is

$$1, \quad -6, \quad -\frac{47}{3}, \quad -52.$$

There is only **one sign change**. Hence, $N_{\text{RHP}} = 1$ Since the polynomial is third order,

$$N_{\text{LHP}} = 3 - 1 = 2.$$

No row of zeros occurs, so $N_{j\omega} = 0$.

Thank You

Questions and Discussion